

Software Requirements Specification (SRS)

Project: Vehicle Parking System (VPS)

Version: 1.0

Authors: Himansh Ahuja (PES2UG24CS923), Shweta Mailaragouda Patil (PES2UG25CS825), Angelo Arakal (PES2UG24CS062), Akarsh Parashuram (PES2UG24CS040)

Date: 09-09-2026 | **Status:** Model Answer / Approved | **Group:** 6

Revision history

Version	Date	Author	Change summary	Approval
1.0	09-09-2026	Himansh Ahuja, Shweta Patil, Angelo Arakal, Akarsh Parashuram	Complete SRS with 18 FRs, 5 NFRs, 4 Security Objectives, 5 Security Requirements, 2 UML Use-Case diagrams, and RTM	Approved

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	Prof. Sheela Devi		
Project Evaluator / Guide	Prof. Pavithra		

1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a **Vehicle Parking System (VPS)**. It defines functional and non-functional requirements, external interfaces, and verification criteria intended for instructors, evaluators, developers, and QA engineers.

1.2 Scope

Covers vehicle entry/exit automation at gate lanes via Automatic Number Plate Recognition (ANPR) and QR tickets, integration with payment gateways for cashless settlement, real-time slot vacancy detection via IoT sensors, multi-category parking allocation, slot pre-booking, operational monitoring, and reporting. Excludes internal payment switch banking implementations and physical towing enforcement.

1.3 Audience

Developers, QA Engineers, System Integrators, Maintenance Technicians, and Assessment Evaluators.

1.4 Definitions

List of acronyms: VPS, ANPR, RFID, QR, POS, UI, API, TLS, PCI-DSS, RTO, RPO, EV, VMS.

2. Overall description

2.1 Product perspective

The VPS is an intelligent self-service and attendant-assisted platform interacting with drivers, attendants, gate hardware (ANPR cameras, boom barriers, loop detectors, ticket printers), overhead slot sensors, and central management servers for transaction settlement and monitoring.

2.2 Major product functions

- Capture vehicle license plates via ANPR camera and issue QR-coded entry tickets.
- Designate nearest vacant parking slot by vehicle type (Two-wheelers, Sedans, SUVs, EVs).
- Detect real-time slot occupancy via sensors and update overhead green/red status LEDs.
- Refresh level-wise availability counts on digital LED boards in real time.

- Compute parking charges dynamically based on duration, tiered hourly rates, and grace periods.
- Process cashless payments at exit via dynamic UPI QR, Credit/Debit cards, or NetBanking.
- Trigger motorized boom barrier release upon verified payment.
- Enable 48-hour advance online slot reservations with no-show auto-cancellation.
- Administrative dashboard for dynamic tariff configuration, audit trails, and financial analytics.

2.3 User roles and characteristics

- **Customer / Driver:** Basic digital proficiency; expects entry/exit in under 3 seconds, intuitive navigation, and fast mobile payments.
- **Parking Attendant:** On-site operator managing exceptions, unreadable plates, cash billing, and manual barrier overrides.
- **System Administrator:** Configures tariffs, user permissions, parking layouts, and monitors system health.
- **Facility Manager:** Reviews revenue reconciliation reports, peak occupancy trends, and security audit logs.

2.4 Operating environment

Embedded gate controller OS (Linux/Windows IoT), secure network (WAN/LAN), power with UPS, environmental protection for outdoor cameras and barrier enclosures, PostgreSQL database, Redis cache.

2.5 Constraints

Wiegand/RS-485 hardware protocols, PCI-DSS payment security compliance, 72-hour autonomous edge offline queuing, cryptographic data encryption at rest and in transit.

3. External interface requirements

3.1 User interfaces

Driver responsive web/mobile app; sunlight-readable kiosk touchscreen with multilingual audio-visual prompts; attendant fast-POS screen; administrative analytics console.

3.2 Hardware interfaces

Motorized boom barriers (relay controller), ANPR IP cameras (RTSP/ONVIF), 2D QR thermal receipt printers, ultrasonic/magnetic slot sensors, variable message LED signs, EMV/NFC payment POS terminals.

3.3 Software interfaces

Payment Gateway API (Razorpay/Stripe over TLS), SMS/WhatsApp notification gateway (Twilio), PostgreSQL relational database, Redis memory store.

3.4 Communications

TLS 1.3 enforced for all client and API channels, MQTT/WebSocket for live slot telemetry, SQLite offline store-and-forward queue on edge gate controllers.

4. System features

Each requirement below includes acceptance criteria and a reference test case. IDs follow VPS-F-####.

4.1 Vehicle Entry & Ticket Generation

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
VPS-F-001	Capture vehicle license plate via ANPR within 1.5 seconds.	Functional	High	Operations	AC-VPS-F-001: Plate OCR $\geq$ 95% accuracy. Test: TC-ENT-01	Requires ANPR camera
VPS-F-002	Issue entry ticket with unique QR code, encrypted session ID, and timestamp.	Functional	High	Customer	AC-VPS-F-002: Ticket issued in $\leq$ 2 seconds. Test: TC-ENT-02	Printer / SMS API
VPS-F-003	Raise entry boom barrier upon valid ticket generation or RFID scan.	Functional	High	Operations	AC-VPS-F-003: Barrier opens in $\leq$ 500ms. Test: TC-ENT-03	Relay controller
VPS-F-004	Prevent barrier descent while vehicle is on safety inductive ground loop.	Functional	High	Safety	AC-VPS-F-004: Halts descent until vehicle clears. Test: TC-ENT-04	Ground loop detector

4.2 Slot Allocation & Real-time Guidance

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
VPS-F-005	Designate nearest vacant slot based on vehicle category (2W, 4W, EV, Handicap).	Functional	High	Driver	AC-VPS-F-005: Matches vehicle class & shortest route. Test: TC-SLOT-01	Allocation engine
VPS-F-006	Update slot status to "Occupied" within 2s of sensor trigger and toggle LED to Red.	Functional	High	Facility	AC-VPS-F-006: DB and LED toggle in $\leq$ 2s. Test: TC-SLOT-02	Ultrasonic sensor
VPS-F-007	Refresh vacancy counts on entrance and floor LED boards with latency $\leq$ 1s.	Functional	Medium	Customer	AC-VPS-F-007: Digital board count matches DB. Test: TC-SLOT-03	VMS display driver

4.3 Automated Fare Calculation & Billing

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
VPS-F-008	Calculate parking fee based on duration, vehicle category, and tiered rate tables.	Functional	High	Finance	AC-VPS-F-008: Fee strictly matches tariff rules. Test: TC-BILL-01	Tariff calculation engine
VPS-F-009	Apply a 15-minute grace period with zero charge for transit drop-offs.	Functional	Medium	Customer	AC-VPS-F-009: Duration $\leq$ 15 min billed at ₹0. Test: TC-BILL-02	Time-rule logic
VPS-F-010	Support dynamic peak-hour surge rates and promo coupon discounts.	Functional	Medium	Business	AC-VPS-F-010: Coupons deduct correctly; surge applies. Test: TC-BILL-03	Coupon & surge engine

4.4 Payment Processing & Exit Management

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
VPS-F-011	Scan ticket QR code or plate at exit to retrieve parking session and invoice.	Functional	High	Customer	AC-VPS-F-011: Invoice retrieved in $\leq$ 1s. Test: TC-PAY-01	Exit scanner / ANPR
VPS-F-012	Facilitate cashless payments via dynamic UPI QR, Credit/Debit cards, and wallets.	Functional	High	Finance	AC-VPS-F-012: Payment webhook marks PAID in $\leq$ 3s. Test: TC-PAY-02	Payment Gateway API
VPS-F-013	Raise exit boom barrier upon verified payment receipt.	Functional	High	Security	AC-VPS-F-013: Barrier opens immediately on PAID status. Test: TC-PAY-03	Exit gate controller
VPS-F-014	Generate itemized GST tax receipt via thermal printer, email, or SMS.	Functional	Medium	Accounting	AC-VPS-F-014: Tax invoice contains GST breakdown. Test: TC-PAY-04	Receipt generator

4.5 Slot Pre-Booking & Reservations

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
VPS-F-015	Allow drivers to reserve a slot up to 48 hours in advance for a specified time window.	Functional	High	Driver	AC-VPS-F-015: Slot locked; reservation code issued. Test: TC-RES-01	Reservation module
VPS-F-016	Automatically release reserved slot if customer fails to arrive within 30 minutes of start time.	Functional	Medium	Operations	AC-VPS-F-016: Slot unlocked on no-show timeout. Test: TC-RES-02	Scheduler service

4.6 Administration, Auditing & Facility Management

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
VPS-F-017	Provide admin console to configure tariffs, zone layouts, and manage attendant credentials.	Functional	High	Admin	AC-VPS-F-017: Tariff updates take effect without restart. Test: TC-ADM-01	Admin portal module
VPS-F-018	Generate exportable daily/monthly revenue, turnaround, and occupancy reports in CSV/PDF.	Functional	Medium	Manager	AC-VPS-F-018: Reports exported in ≤ 5 seconds. Test: TC-ADM-02	Analytics engine

5. Non-functional requirements

NFRs below are measurable and tied to test plans. IDs follow VPS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
VPS-NF-001	Overall vehicle ingress transaction response (from plate scan to ticket issue and barrier raise) shall be ≤ 3 seconds for 90% of transactions under normal load.	Performance	High	90th percentile ≤ 3 s in production-like test. Test: TC-Perf-01
VPS-NF-002	System shall provide 99.9% availability monthly; scheduled maintenance windows excluded.	Reliability	High	Uptime reports show $\geq 99.9\%$ per month. Test: TC-Rel-01
VPS-NF-003	System shall scale to monitor up to 10,000 parking slots and 50 concurrent gate lanes without latency exceeding 200ms.	Scalability	High	API latency remains < 200 ms at 500 req/sec. Test: TC-Scale-01
VPS-NF-004	All sensitive cardholder and customer PII data must comply with PCI-DSS; card details and plain credentials must not be stored in plaintext.	Security / Compliance	High	PCI-DSS audit checklist pass; no plaintext data. Test: TC-Sec-01
VPS-NF-005	Edge gate controllers shall provide autonomous offline transaction queuing for up to 72 hours during wide-area network failure.	Fault Tolerance / Resilience	High	Lane functions continuously during simulated WAN disconnection; auto-syncs in ≤ 5 min on reconnect. Test: TC-Fault-01

5.1 Security

5.1.1 Security Objectives

This section lists 4 core security objectives for the project:

1. **Confidentiality & Data Protection:** Safeguard customer contact information and vehicle license plate movement logs from unauthorized access or interception.
2. **Transactional Integrity & Anti-Fraud:** Prevent financial revenue loss, counterfeit ticket issuance, and unauthorized manual barrier opening through cryptographically signed tickets and immutable audit records.
3. **System Availability & Perimeter Defense:** Protect ingress/egress endpoints and API gateways against Distributed Denial of Service (DDoS), SQL injection, and sensor replay attacks.
4. **Access Accountability & Non-Repudiation:** Ensure every administrative configuration, shift cash declaration, and gate override is authenticated, authorized, and logged with operator identity and timestamp.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance criteria / Test case ref
VPS-SR-001	TLS 1.3+ mandatory for all network connections (web apps, kiosks, APIs, and sensors).	Security	High	Unencrypted connections rejected; Qualys SSL Grade A+. Test: TC-SEC-01
VPS-SR-002	Hash all passwords with Argon2id/bcrypt and encrypt sensitive records at rest using AES-256.	Security	High	Plaintext passwords and PII absent from database dumps. Test: TC-SEC-02
VPS-SR-003	Enforce Role-Based Access Control (RBAC) and Multi-Factor Authentication (MFA) for administrative accounts.	Security	High	Admin login requires TOTP; non-admin blocked. Test: TC-SEC-03
VPS-SR-004	Digitally sign entry ticket QR codes with HMAC-SHA256 to prevent counterfeit tickets.	Security	High	Tampered QR tickets rejected with security alert. Test: TC-SEC-04
VPS-SR-005	Maintain immutable, write-once audit logs for all barrier overrides and tariff edits, retained for 3 years.	Security / Audit	High	Log tampering detected; audit trail complete. Test: TC-SEC-05

6. Quality attributes & Acceptance tests

- **Exit criteria for acceptance:** All 18 high-priority functional requirements implemented and verified, no critical NFR failures, zero high security vulnerabilities, and RTM shows all test cases passed.
- **Acceptance test suites:** Ingress Automation, Slot Allocation, Dynamic Billing, Cashless Payments, Reservation Lifecycle, Performance/Scale, Security & Audit, and Network Resilience.

7. System models and diagrams

7.1 UML Use-Case Diagram 1: Vehicle Ingress, Parking & Egress Operations

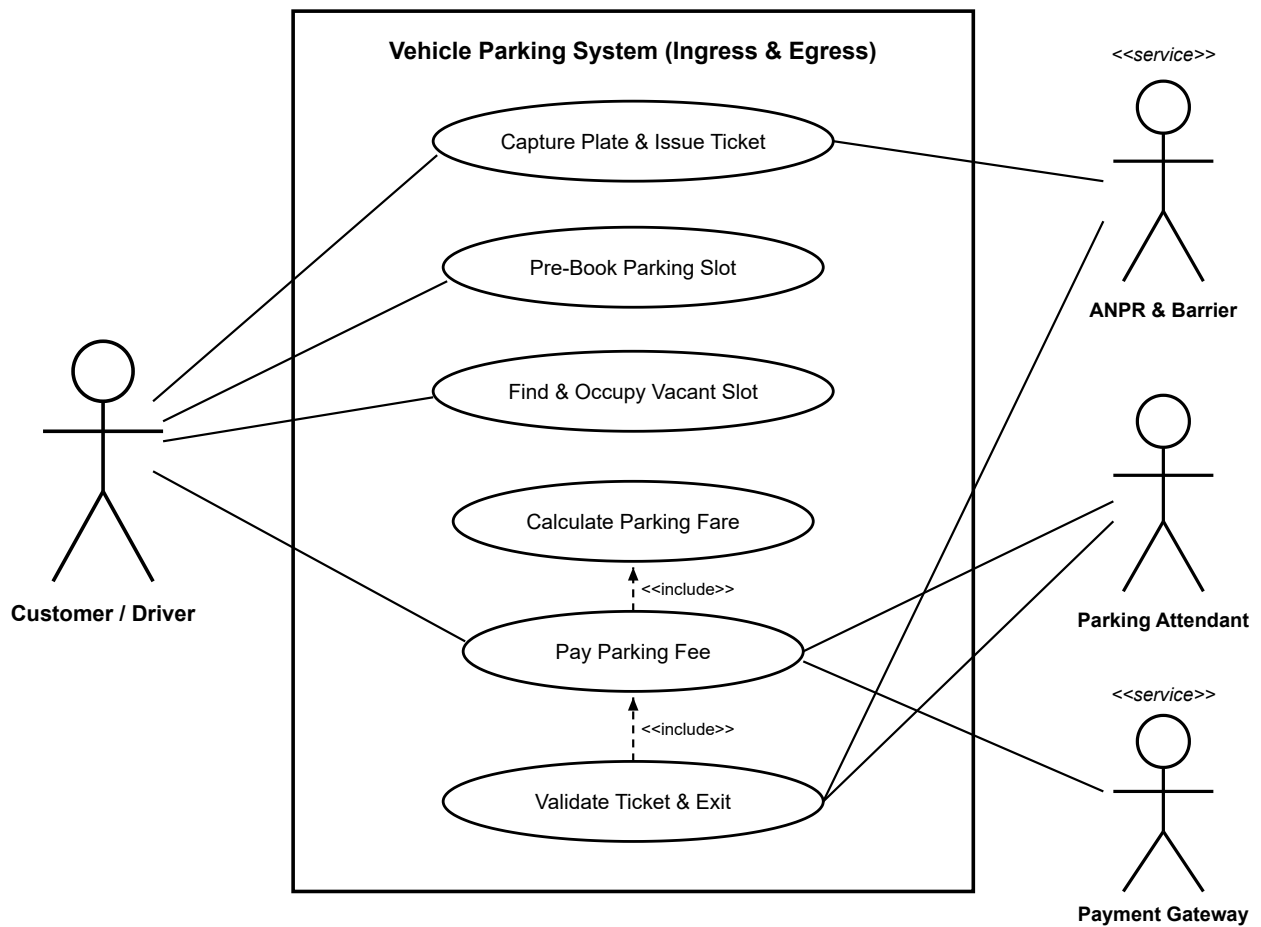


Figure 7.1: UML Use-Case Diagram for Vehicle Ingress, Parking, and Egress Operations

7.2 UML Use-Case Diagram 2: Administration & Facility Management

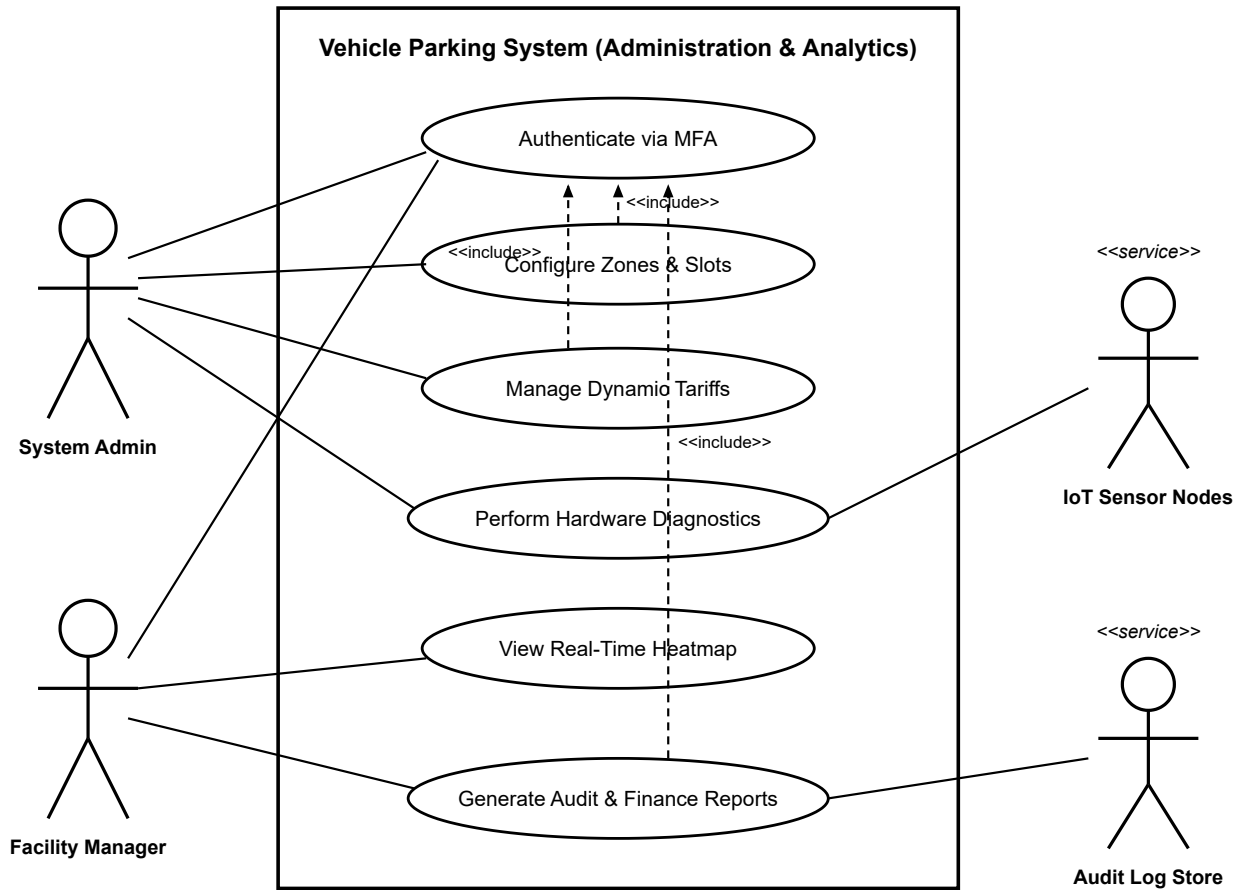


Figure 7.2: UML Use-Case Diagram for System Administration and Analytics

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status*	Comments
VPS-F-001	ANPR Plate Capture (≤ 1.5s)	4.1 / DS-ENT-01	IngressModule	TC-ENT-01	P	Verified with IR camera
VPS-F-002	Ticket / QR Generation	4.1 / DS-ENT-02	IngressModule	TC-ENT-02	P	Thermal printer & SMS
VPS-F-003	Ingress Barrier Raise Signal	4.1 / DS-ENT-03	GateController	TC-ENT-03	P	Relay actuation verified
VPS-F-004	Ground Loop Safety Check	4.1 / DS-ENT-04	SafetyModule	TC-ENT-04	P	Anti-tailgating loop
VPS-F-005	Nearest Vacant Slot Assignment	4.2 / DS-SLOT-01	AllocationEngine	TC-SLOT-01	P	Category routing validated
VPS-F-006	Real-Time Occupancy LED Switch	4.2 / DS-SLOT-02	SensorHub	TC-SLOT-02	P	Sub-2s LED switch
VPS-F-007	Entrance VMS Display Refresh	4.2 / DS-SLOT-03	SignageService	TC-SLOT-03	P	MQTT payload tested
VPS-F-008	Dynamic Tiered Fare Computation	4.3 / DS-BILL-01	TariffEngine	TC-BILL-01	P	Time billing verified
VPS-F-009	15-Minute Grace Period Zero Charge	4.3 / DS-BILL-02	TariffEngine	TC-BILL-02	P	Boundary values verified
VPS-F-010	Peak Surge & Promo Discounts	4.3 / DS-BILL-03	TariffEngine	TC-BILL-03	P	Coupon formulas verified
VPS-F-011	Exit QR / Plate Scanning	4.4 / DS-PAY-01	EgressModule	TC-PAY-01	P	OCR & QR fetch tested
VPS-F-012	Cashless Payment Processing	4.4 / DS-PAY-02	PaymentGateway	TC-PAY-02	P	UPI & Card webhooks
VPS-F-013	Egress Barrier Release	4.4 / DS-PAY-03	GateController	TC-PAY-03	P	Barrier raises on PAID
VPS-F-014	Tax Receipt / SMS Dispatch	4.4 / DS-PAY-04	ReceiptService	TC-PAY-04	P	PDF/SMS receipts
VPS-F-015	48h Advance Slot Pre-Booking	4.5 / DS-RES-01	BookingModule	TC-RES-01	P	Advance slot lock
VPS-F-016	30-Minute No-Show Auto-Release	4.5 / DS-RES-02	BookingModule	TC-RES-02	P	Timeout cron verified
VPS-F-017	Admin Dashboard & Rate Config	4.6 / DS-ADM-01	AdminPortal	TC-ADM-01	P	Live tariff updates
VPS-F-018	Analytics & Exportable Reports	4.6 / DS-ADM-02	AnalyticsEngine	TC-ADM-02	P	CSV/PDF reports
VPS-NF-001	Total Ingress Latency ≤ 3s	5.1 / DS-PERF-01	Whole System	TC-Perf-01	P	Load benchmark passed

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status*	Comments
VPS-NF-002	99.9% Monthly Availability	5.1 / DS-REL-01	DevOps / Infra	TC-Rel-01	P	Uptime checks verified
VPS-NF-003	Scalability (10k slots, 50 lanes)	5.1 / DS-SCALE-01	Database / Redis	TC-Scale-01	P	Concurrent load tested
VPS-NF-004	PCI-DSS Compliance & PII Encryption	5.1 / DS-SEC-01	PersistenceLayer	TC-Sec-01	P	PCI-DSS audit passed
VPS-NF-005	72h Edge Gate Offline Resilience	5.1 / DS-FLT-01	EdgeController	TC-Fault-01	P	Offline queue sync
VPS-SR-001	Mandatory TLS 1.3+ Encryption	5.2 / DS-SEC-01	ReverseProxy	TC-SEC-01	P	Cipher suites verified
VPS-SR-002	AES-256 Storage & Argon2id Hash	5.2 / DS-SEC-02	PersistenceLayer	TC-SEC-02	P	Data encryption verified
VPS-SR-003	RBAC & Admin MFA Authentication	5.2 / DS-SEC-03	AuthModule	TC-SEC-03	P	TOTP validation passed
VPS-SR-004	HMAC-SHA256 Signed QR Tickets	5.2 / DS-SEC-04	IngressModule	TC-SEC-04	P	Tampered QR rejected
VPS-SR-005	Immutable Audit Log Retained 3 Yrs	5.2 / DS-SEC-05	AuditLogger	TC-SEC-05	P	Tamper-evident logs

* Status Key: N = Not Started, P = Passed / Verified, A = Approved